Solving by Elimination Method

This is another algebraic method for solving a pair of linear equations. This method is more convenient than the substitution method. Here we eliminate (i.e. remove) one of the two variables in a pair of linear equations, so as to get a linear equation in one variable which can be solved easily.

The various steps involved in the technique are given below:

Step 1: Multiply one or both of the equations by a suitable number(s) so that either the coefficients of first variable or the coefficients of second variable in both the equations become numerically equal.

Step 2: Add both the equations or subtract one equation from the other, as obtained in step 1, so that the terms with equal numerical coefficients cancel mutually.

Step 3: Solve the resulting equation to find the value of one of the unknowns.

Step 4: Substitute this value in any of the two given equations and find the value of the other unknown.

Example 3.50

Given $4a + 3b = 65$ and $a + 2b = 35$ solve by elimination method.

Solution

Given, $4a + 3b = 65$ .....(1)

$a + 2b = 35$ .....(2)

(2) × 4 gives $4a + 8b = 140$

(–) (–) (–)

Already (1) is $\underline{4a + 3b = 65}$

$5b = 75$ which gives $b = 15$

Put $b = 15$ in (2):

$a + 2(15) = 35$ which simplifies to $a = 5$

Thus the solution is $a = 5$, $b = 15$.

Verification :

$4a+3b = 65$ ...(1)

$4(5)+3(15) = 65$

$20 + 45 = 65$

$65 = 65$ True

$a + 2b = 35$ ...(2)

$5 + 2(15) = 35$

$5+30 = 35$

$35 = 35$ True

Example 3.51

Solve for x and y: $8x - 3y = 5xy$, $6x - 5y = -2xy$ by the method of elimination.

Solution

The given system of equations are $8x - 3y = 5xy$ ...(1)

$6x - 5y = -2xy$ ...(2)

Observe that the given system is not linear because of the occurrence of xy term. Also note that if x =0, then y =0 and vice versa. So, (0,0) is a solution for the system and any other solution would have both $x \uparrow 0$ and $y \uparrow 0$.

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